Cal State Fullerton
College of Engineering and Computer Science

CPSC 483: *Introduction to Machine Learning*

Fall 2026

Prerequisites:

- CPSC 375, MATH 237 or CPSC 335; MATH 338, PSYC 201, PUBH 349, ISDS 210 or EGEC 323
- Computer Science or Computer Engineering major or minor; or Computer Science or Computer Engineering graduate standing

Course	Room	Class Time	Final Examination
CPSC 483-01	Hum 226	Lecture: TuTh 5:30PM - 6:45PM	Dec 17, 5:00PM - 6:50PM

INSTRUCTOR: Anli Ji
OFFICE: CS-538
EMAIL: anli.ji@fullerton.edu (best way to reach me)
PHONE: (657)278-2871
ZOOM LINK: https://fullerton.zoom.us/j/6532946983
OFFICE HOUR: Tuesday and Thursday 9:30 am – 11:00 am or by appointment.

Response time

If you are sending email to the instructor, please include course title "**CPSC 483**" at the beginning of the subject. You should always send email messages from your **fullerton.edu** email account. There will be no reply to any free web-based email accounts such as Gmail, Hotmail or Yahoo. The instructor will strive to respond to email questions, phone calls, and online assignments in one-to-two business days (excludes weekends or holidays).

Course Communication

All course announcements and individual emails are sent through CANVAS which only uses CSUF email accounts. Therefore, you MUST check your CSUF email on a regular basis (several times a week) for the duration of the course.

Technical Problems

If you encounter any technical difficulties, contact the instructor immediately to document the problem.
Contact: student IT help desk, email, phone: (657)278-8888, walk-in student genius center, online chat - log into portal; click "Online IT Help"; click "Live Chat."
For issues with Canvas: Canvas Support = 657-278-8888, search the CSUF Canvas Guides with AI Assistant, report a technical problem.

In case of technical difficulties with CANVAS or other online resources, your instructor will communicate with students directly through their CSUF email. The instructor will provide directions on alternative methods for submitting work and/or may extend submission deadlines. If email is not available, students may contact the department office for guidance.

Course Catalog Description
Design, implement and analyze machine learning algorithms, including supervised learning and unsupervised learning algorithms. Methods to address uncertainty. Projects with real-world data.

Student Learning Objectives
This course will provide students with the fundamental principles and techniques of machine learning. The learning objectives are listed as following:

- Acquire fundamental knowledge about machine learning concepts, algorithms, and applications
- Be able to prepare and preprocess data for machine learning tasks
- Understand the concepts and techniques of supervised and unsupervised learning
- Be able to apply common machine learning algorithms to solve classification, regression, and clustering problems
- Be able to evaluate and compare machine learning models using appropriate performance metrics and validation techniques
- Demonstrate the ability to implement machine learning solutions using Python and commonly used machine learning libraries
- Understand common machine learning challenges, including overfitting, underfitting, model generalization, and feature selection

Course Materials
Required Text: *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 2nd Edition
Lecture notes will be uploaded within 24 to 48 hours after the class

Optional

- *A First Course in Machine Learning*, Rogers and Girolami, 2nd Ed., Chapman & Hall/CRC, ISBN: 1498738486
- *The Elements of Statistical Learning: Data mining, Inference, and Prediction*, T. Hastie, R. Tibshriani, and J. Friedman, 2nd edition, Springer, 2009.
- *Machine learning: A probabilistic perspectiv*e, Kevin Murphy, MIT press, 2012.
- *Machine learning*, Tom Mitchell, McGraw-Hill, 1997. [Free PDF]
- *Understanding Machine Learning from theory to Algorithms*, Shai Shalev-Shwartz and Shai Ben-David, Cambridge University Press, 2014 [Download for personal use]
- *A Whirlwind Tour of Python*, Jake VanderPlas, O'Reilly, ISBN: 978-1-491-96465-1

Exams
There will be one midterm and one final exam given for this course, both administered in person. The exams are NOT cumulative. Midterm will cover materials from the first half of the semester (week 1-8) and final will cover from after the midterm to the end of the semester. Students are permitted to use a single double-sided cheat sheet during each exam. No external sources (e.g., cellphones, laptops, computers, etc.) nor collaboration are allowed on any exam.

Students are expected to arrive on time for all exams. Make-up exams will NOT be offered except in cases of extraordinary circumstances (e.g., serious illness or accident), and only if appropriate documentation can be provided and prior notice is given ahead. Keep all assignments and exams returned to you so that any discrepancies can be easily and fairly resolved. **Requests for re-evaluation of graded material must be made within one week of grades being posted.**

Students MUST take examinations in the section for which they are officially enrolled. Taking an exam in a different section is not permitted and will result in a grade of zero for that exam. This policy ensures fairness and consistency, as exam content, format, and scheduling may differ between sections.

Homework Assignments

There will be approximately four homework assignments during the semester. All assignments must be submitted electronically via Canvas by the specified due date. Collaboration among students is permitted; however, each student must submit their own work, and **plagiarism will not be tolerated**.

Questions or concerns regarding grading must be submitted in writing by email, with a clear explanation, within one week of the grade being posted. Late work will not be accepted unless the student provides compelling reasons with appropriate documentation prior to the deadline and receives approval from the instructor.

Late Submission

All assignments are due on the date and time specified. A one-day grace period is allowed, during which assignments may still be submitted without penalty. Any work submitted after the one-day grace period will NOT be accepted and will receive a grade of zero for that assignment.

Group Projects

There will be one group project for groups of 1-5 people. The project will involve the design, implementation, and analysis of a substantial database system.

- Project Timeline: The project will include approximately two submissions, spaced about 7 weeks apart. A detailed project overview and requirements will be provided during the second week of instruction.
- Collaboration: Students are expected to dedicate time outside of class for project collaboration.
- In-Class Checkpoints: Portions of class time will be used for progress reviews, during which each group will present updates and receive feedback.

Survey paper

Graduate students enrolled in the course are required to write an extra assignment of survey paper (UPS 411.100), while for undergraduate students the survey paper is optional and may be completed for **extra credit**. The topic must be directly related to the course content and approved by the instructor in advance. The paper must be submitted through Canvas no later than Friday of Week 15 at 11:59 p.m. The paper should follow the IEEE two-column format in 11-point font, with a minimum length of three pages (excluding references).

Grading Policies and Standards

Undergraduate			***Graduate***	
	Percentage			***Percentage***
Homeworks	15%		Homeworks	15%
Midterm Exam	25%		Midterm Exam	25%
Final Exam	30%		Final Exam	30%
Project	30%		Project	25%
Survey Paper (EC)	5%		Survey Paper	5%

Grading scale

Grade	*Percentage*		*Grade*	*Percentage*
A+ = 4.0	98 - 100%		C+ = 2.3	77 - 79.9%
A = 4.0	93 - 97.9%		C = 2.0	73 - 76.9%
A- = 3.7	90 - 92.9%		C- = 1.7	70 - 72.9%
B+ = 3.3	87 - 89.9%		D+ = 1.3	67 - 69.9%
B = 3.0	83 - 86.9%		D = 1.0	63 - 66.9%
B- = 2.7	80 - 82.9%		D- = 0.7	60 - 62.9%
			F = 0.0	0 - 59.9%

Curving-up/down may be considered based on a comparison between the class average GPA and the university-wide average GPA

Academic Integrity

It is expected that any work submitted for a grade in this course will be the sole product of the individual student, unless otherwise permitted. Presenting work from other sources as your own is unacceptable and will result in a notification to the Dean of Students of a violation of campus standards. Behaviors that seek to gain an unfair advantage or negatively impact the ability of others to learn will also result in a report to the Dean of Students. The consequences for cheating or other actions contrary to CSUF student conduct polices can be severe. Communicate with your instructor if you find yourself in a situation that might lead to a poor decision.

You are encouraged to use the resources provided by your instructor. Discuss the use of other online resources with your instructor. The use of sites, including but not limited to Chegg and Course Hero, which require subscriptions and provide solutions to homework problems, exam questions, etc., is explicitly prohibited in this course and is considered academic dishonesty.

Collaborating has been made easier by the many tools available to use on the internet (e.g. Discord, Zoom, Microsoft Teams). I encourage you to use these tools to work together, to form study groups, etc. However, any sharing of assignments (even if only intended to help) or using these communication tools for unauthorized collaboration is considered academic dishonesty. Unless otherwise explicitly stated by the instructor, assignments, and examinations must be completed on your own.

ChatGPT and Other LLMs

Generative AI tools, such as ChatGPT, Claude, Gemini, GitHub Copilot, and other Large Language Models (LLMs), are increasingly used in the technology. These tools can be valuable resources, but they are NOT always accurate or reliable. AI-generated responses may contain incorrect information, inappropriate assumptions, security issues, or code that does not work as intended. An important goal of this course is to help you develop the knowledge and skills necessary to become a reliable and independent computing professional.

You may use generative AI tools to support your learning, such as:

- Asking for additional explanations of concepts discussed in class.
- Reviewing terminology or background information.
- Generating examples for practice or self-study.

- Asking questions about errors or concepts after making your own attempt.
- Exploring alternative approaches to a problem for learning purposes.

Unless explicitly permitted by the instructor for a particular assignment, you **may NOT use generative AI tools** to:

- Generate answers for quizzes, exams, assignments, or other graded assessments.
- Generate code or written responses that you submit as your own work.
- Complete substantial portions of an assignment on your behalf.
- Rewrite AI-generated work and submit it as if it were independently produced.
- Use AI during an assessment where outside assistance is not permitted.

You are expected to complete graded work based on your own understanding of the course material and the skills developed through lectures, activities, readings, and practice.

Policy on the use of generative AI or other technology: https://fdc.fullerton.edu/teaching/ai.html

Student Resources Website

See student information for course syllabi for details and explanations of the University policy areas:

- University learning goals
- General Education learning objectives
- Netiquette/appropriate online behavior
- Students' rights to accommodations
- Campus student support resources
- Academic integrity
- Emergency preparedness/what to do
- Library services
- Student IT services and competencies
- Software privacy and accessibility
- Accessibility statement
- Diversity statement
- Land acknowledgement
- Final exam schedule
- Semester calendar

Please inform the instructor during the first week of classes about any disability or special needs that you may have that may require specific arrangements related to attending class sessions, carrying out class assignments, or writing papers or examinations. Any student who, because of a disability, may require special arrangements in order to meet course requirements must register with the Office of Disability Support Services within the first week of classes.

The Office of Disability Support Services' website is http://www.fullerton.edu/DSS/ . They can be reached by phone at 657-278-3117, TDD at 657-278-2786, or email at dsservices@fullerton.edu . Their office is located in University Hall, room 101. The instructor may request verification of need from the Dean of Students Office.

Classroom Management

Cell phones. Out of courtesy and respect for others, you should turn off or silence your cell phone during class.

In-class Recording. Requests to record class meetings (outside of accommodation requirements) will be decided on a case-by-case basis by the instructor. If permitted, any approved recordings are for private use during the semester for the purposes of studying and "shall not be made publicly accessible without the written consent of the instructor and any students recorded in the class" (UPS 330.230). Any approved recordings must be deleted at the end of the semester.

Technology. If you need any assistance with technology, including checking out a laptop computer for the semester, obtaining ancillaries (e.g., webcam, microphone), accessing mi-fi, downloading free software for class or personal use, or other help, you can find what you need at the student technology services website. Another useful website is IT essential resources.

Tentative Schedule of Classes

Week	***List of Topics (subject to change)***	***Chapters***	***Assignments Due***
1 (Aug 25, 27)	Welcome, Introduction to Machine Learning	Syllabus; 1	
2 (Sep 1, 3)	Data Preprocessing Group Project Review	2	Initial Group Form
3 (Sep 8, 10)	Classification and Performance Measures	3	HW 1 posted
4 (Sep 15, 17)	Linear Regression, Logistic Regression	4	HW 1 due
5 (Sep 22, 24)	Support Vector Machines	5	
6 (Sep 29, Oct 1)	Decision trees	6	HW 2 posted
7 (Oct 6, 8)	Ensemble Learning	7	HW 2 due
8 (Oct 13, 15)	Review for midterm Midterm Exam		
9 (Oct 20, 22)	Dimensionality Reduction	8	Group Project Update
10 (Oct 27, 29)	Unsupervised Learning (Clustering, Gaussian Mixture Models)	9	HW 3 posted
11 (Nov 3, 5)	Artificial Neural Networks (Multilayer perceptrons)	10	HW 3 due
12 (Nov 10, 12)	Artificial Neural Networks (CNN)	14	HW 4 posted
13 (Nov 17, 19)	Recurrent Neural Network	15	HW 4 due
14	Fall Recess – NO CLASSES		
15 (Dec 1, 3)	Reinforcement Learning	18	Group Project due
16 (Dec 8, 10)	Project Presentations Review for final exam		
17	Final Exam		